Derek Dietz

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EDUCATION

Northwestern University, *MS in Robotics* Aug 2026
College of William & Mary, *BS in Physics* May 2022

SKILLS

Software: Python, C++, C, Git, Linux, Docker, CMake, Bash, Matlab, PyTorch, 3DGS, NeRF
Robotics: ROS/ROS2, PX4, MAVROS, ArduSub, SLAM, MoveIt, tf2, Gazebo, Rviz2, MuJoCo, Mechatronics
Automation: PID Control, Motion Control, Brushed DC Motors, Sensor Integration
Machine Learning: Supervised, Unsupervised, MLP, CNN, YOLO, Reinforcement Learning, GAN

EXPERIENCE

Epic Systems - Madison, WI, *Technical Solutions Engineer* Sept 2023 – May 2025
- Developed custom software solutions to meet client integration requirements on a large safety-critical platform
- Diagnosed and resolved integration failures in live production environments across multiple system layers
- Collaborated with engineering and client teams to gather requirements and document solutions for maintenance

Ascend Engineering - Chicago, IL, *Autonomy Intern* June 2026 – Present
- Custom development in PX4 for various UAV autonomy tasks
- Developed ROS2 packages for autonomy tasks requiring real time sensor integration

PROJECTS

Sensing and Grasping with Franka Arm (ROS2, Python, MoveIt, YOLO) November 2025
- Developed a full manipulation behavior pipeline on a 6-DOF Franka arm from perception through task execution
- Implemented motion planning, collision checking, and scene management using the ROS2 MoveIt stack
- Integrated Intel RealSense D435i with a YOLO model for real-time object detection and pose estimation
- Resolved a pose estimation bottleneck via multi-angle measurement aggregation, enabling reliable grasping

Sensing and Retrieval using BlueROV2 (ROS2, Python, OpenCV, YOLOv8) March 2026
- Designed and validated autonomous behavior execution logic for underwater object retrieval on a physical ROV
- Built a modular ROS2 architecture cleanly separating perception, control, and actuation layers
- Implemented closed-loop PID control integrating vision-based detection with autonomous grab sequencing
- Trained and deployed a custom YOLOv8n model in ROS2 for real-time underwater object detection

Extended Kalman Filter SLAM (C++, ROS2, CMake, Catch2) March 2026
- Developed a modular EKF SLAM system in with clean separation across odometry, detection, and estimation
- Built a C++ rigid body transformation library validated with an extensive Catch2 unit test suite via CMake
- Implemented circle detection and data association to extract and track landmarks from laser scan data

Robotic Mapping with 3DGS (Python, C++, 3DGS) March 2026
- Developing pipeline for depth uncertainty estimation using 3DGS on Hello Robot Stretch 3
- Implementing autonomous path planning algorithm using custom trained 3DGS model

AWARDS

American Institute for Aeronautics and Astronautics (AIAA) January 2023
- 1st place winner of the SciTech Idea Challenge

Virginia Microelectronics Consortium (VMEC) August 2021
- Gold Award winner for research presentation on photolithography toppling angles